



Pandas can neutralise Cyanide

A recent study revealed that Pandas can ingest over 65 percent of harmful cyanide from bamboo shoots and chill like it's nothing. <http://digit.in/PandaCyanide>



Robo-fish

Scientists have created a soft robotic fish called So-Fi to learn about hard-to-find ocean creatures <http://digit.in/robofish>

GOING WITH THE FLOW

Understand the giant oceanic conveyor belts of the Earth

Arbab Mukherjee | arnab@digit.in

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uch happens in the waters of this world that affects our lives on land. Beyond natural disasters like

tsunamis and cyclones, there are some pretty regular phenomenon happening in the world's oceans that continuously alter and influence our everyday lives, regardless of where we are on the planet. One such phenomenon is ocean currents.

Broadly speaking, an ocean current is any regularly occurring mass movement of water, primarily horizontal, in all of the world's major water bodies. But they're much more than just good old H₂O ambling about the globe. Much of the world's climate and biodiversity has been affected and determined by ocean currents. How exactly? Let's find out.

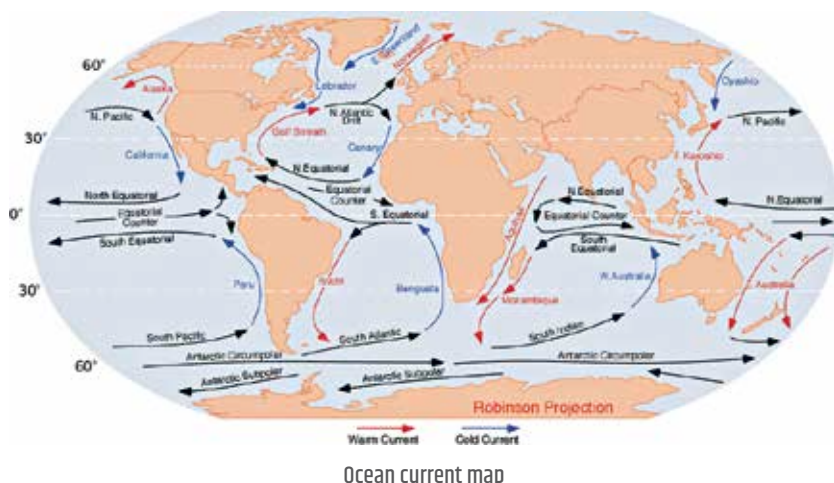
WHAT CREATES OCEAN CURRENTS

To understand why ocean currents are what they are, we need to grasp how they come to be. There are a multitude of factors that influence the formation and propagation of ocean currents. But before we get into them, our understanding of ocean currents needs to be

further subdivided into its two major types - surface currents and deep water currents.

Surface currents form about 10% of the water in the oceans. The rest of the 90% of the oceans are comprised of the thermohaline currents in deep water. Here "thermo" means temperature and "haline" means salinity, which gives us the first two factors that affect cur-

Solar heating, especially around the equators, leads to faster evaporation and a higher density of salt in the water. It also causes expansion in water volume, leading to a water level higher by 8cm near the equator as compared to mid-latitudes. Along with that, persistent surface winds lead to friction between the layers of water, with the upper layers having a greater



rent formation. Essentially, different temperatures and salinity give rise to different densities by depth. As colder, more saline water sinks deeper, warmer water seeks to replace it which gives rise to massive movements of water. This sinking happens mostly towards the poles where the water cools faster and ice formation leaves behind more salt. But this creates a different picture of ocean currents than what happens in reality - which means that there are more factors affecting it.

tendency to flow in the direction the wind is blowing as compared to the lower layers.

On top of these factors, there's the coriolis effect to account for. Without going into much detail, let us tell you that the coriolis force is the force that makes objects maintain the rotational inertia that earth's rotation grants them. So in the currents moving towards the poles, it acts towards the right or left depending on which hemisphere you're in. All these factors